

9a	The Magnetic Flux Linkage in a coil is the product of the area of the coil and the number of turns in the coil and the component of the magnetic flux density perpendicular to that area.	E	M1 A1
9b	$t = 1.2 \text{ s}$ $\text{e.m.f} = \frac{\Delta N B A}{t} = \frac{540 \times 0.250 \times \pi \times 0.030^2}{1.2}$ $= 0.32 \text{ V}$	A	B1 C1 A1
9ci	light damping	E	A1
9cii	sheet cuts (magnetic) flux and causes induced emf (induced) emf causes (eddy) currents (in sheet) either currents (in sheet) cause resistive force or currents (in sheet) dissipate energy <u>smaller currents</u> in Y or <u>larger</u> currents in X, so dashed line is X	D	B1 B1 B1 B1
9di1	sinusoidal waves with amplitude 320 V and two time periods shown, each of 0.020 s $\omega = \frac{2\pi}{T}$ $T = \frac{2\pi}{100\pi}$ $= 0.020 \text{ s}$	A	B1 B1
9di2	If 9di1 is positive sine function shown, sinusoidal half waves in positive V only for the first half of cycle and (peak at 320 V, line at zero for second half of cycle (two time periods shown, each of 0.020 s) If 9di1 is negative sine function shown, then vice versa – line at zero for first half cycle, then sinusoidal half waves in negative V only for second half cycle.	D	B1 B1
9di3	If 9di1 is positive sine function shown, line at <u>zero for first half</u> of cycle and <u>sinusoidal half waves in negative V only for the second half</u> of cycle. (peak at 320 V two time periods shown, each of 0.020 s) If 9di1 is negative sine function shown, then vice versa – sinusoidal half waves in negative V only for first half cycle, then line at zero for second half cycle.	D	B1
9dii	Average power $P = \frac{1}{2} (V_{\text{rms}}^2/R)$ Or $\frac{1}{2}(P_o/2)$ $= [\frac{1}{2} (\frac{320}{\sqrt{2}})^2]/120$	A	C1

	$= 210 \text{ (213) W}$ Or Average power $P = \frac{1}{2}(P_o/2)$		A1
9diii	The root-mean-square p.d is defined as <u>the steady direct current</u> which produces <u>heat at the same rate</u> as the alternating current across a given resistor.	E	B1
9div	Average power $= I_{\text{rms}}^2 R$ $I_{\text{rms}} = \sqrt{\frac{213}{120}}$ $= 1.33 \text{ A}$ Accept ecf Or $I_{\text{rms}} = \frac{V_{\text{rms}}}{R} = \frac{V_o}{2R}$	A	C1 A1